

UNIVERSITY OF SOUTHERN CALIFORNIA

Statistical genetics on All of Us

A first workflow in the VM terminal

Jeffrey M. Swigert, PhD

Center for Economic and Social Research

September 2026

40 minutes of instruction 20 minutes of practice

What you will leave with

- A working terminal workflow you can rerun.
- A QC report showing which samples and variants remain.
- An association result you can explain, including its limitations.
- A saved run with commands, logs, and input provenance.

0–12 min	Workspace, VM, terminal, and cloud files
12–27 min	Genotypes, sample IDs, and QC
27–35 min	Covariates, regression, and interpretation
35–40 min	Saving, disclosure rules, and the sandbox
40–60 min	Your own complete run

The analysis we will build

Step	What must be true before moving on
1. Locate inputs	Release, format, genome build, and file paths are known.
2. Match people	Phenotypes and covariates join to genotypes by unique IDs.
3. Apply QC	Every exclusion has a reason and a recorded count.
4. Fit a model	Covariates and relatedness strategy fit the study.
5. Read and save	Alleles, sample size, uncertainty, and provenance are clear.

A successful command is one checkpoint in a scientific workflow.

Where things live in All of Us

Place	What it contains	How we use it
CDR tables	Phenotypes and other structured records	Query with SQL / BigQuery
Genomic resources	Genotype files and QC meta-data	Choose the correct callset
VM disk	Working copies and running processes	Run PLINK 2 in a terminal
Workspace bucket	Files saved for the workspace	Preserve inputs and runs

The terminal runs on the cloud VM. Your browser is the interface.

Sources: AoU Data Dictionaries; Verily Workbench JupyterLab documentation. Links in appendix.

The workspace version matters

Updated Workbench

CDR v9 is available here.

Use the selected data collection's reference resources.

Copy the bucket URI from the workspace Resources tab.

Older scripts

May assume `WORKSPACE_CDR`, `WORKSPACE_BUCKET`, or `GOOGLE_PROJECT`. Check each value against the actual workspace before reuse.

Our settings: `config.sh` names the input prefix and output bucket explicitly.

Sources: AoU CDR v9 announcement; Data Dictionaries; Verily JupyterLab. Checked 18 September 2026.

A few terminal commands go a long way

Open your running JupyterLab app, then a **Terminal** from the Launcher.

```
pwd                # current directory
ls -lh             # files and sizes
cd aou_cli_workshop # workshop folder
head -n 5 data/toy.pvar # preview text metadata
less scripts/02_qc.sh  # read code; q exits
bash scripts/00_preflight.sh # check the environment
```

Useful habits

Tab completes names. Up-arrow recalls a command. Ctrl+C interrupts a running command.

Reading connection: Mills, Barban, and Tropf (2020), Chapter 8, section 8.2.

Local paths and cloud addresses

```
# Local VM file
aou_cli_workshop/data/toy.pgen

# Cloud object, with the real URI supplied by the instructor
gcloud storage ls "$WORKSHOP_BUCKET/statgen-workshop/"

# Copy a SMALL prepared file from a workspace bucket
gcloud storage cp "$WORKSHOP_BUCKET/training/README.txt" .
```

Working rule: run PLINK on small files staged on the VM, then save the run to your workspace bucket.

Source: Verily Workbench JupyterLab storage guidance. Bucket paths above are illustrative.

Genotypes arrive in several formats

Format	Contents	This workshop
VCF	Variants, alleles, per-sample calls	Synthetic input generation
PLINK 1	.bed, .bim, .fam	Recognize existing data
PLINK 2	.pgen, .pvar, .psam	Main working format
Hail MT / VDS	Dense matrix / sparse variant data	Later, for larger preparation

One prefix, three files:

`toy.pgen` holds calls; `toy.pvar` identifies variants; `toy.psam` identifies samples.

Sources: supplied Chapter 7, section 7.5; PLINK 2 file-format reference; AoU genomic data organization.

A genotype becomes an allele count

At one biallelic autosomal site with REF = A and ALT = G:

Genotype	VCF GT	ALT count	Meaning
A/A	0/0	0	No G copies
A/G	0/1	1	One G copy
G/G	1/1	2	Two G copies
Unknown	./.	Missing	Exclude from that test

Dosage can be fractional when it represents an expected allele count.

Always name the counted allele. REF, ALT, and minor allele have different meanings.

Matching samples is a statistical operation

Genotype metadata

IID	File position
SIM0003	1
SIM0001	2
SIM0002	3

Phenotype table

IID	trait
SIM0001	1.7
SIM0002	3.2
SIM0003	2.1

- Join by the sample key; verify uniqueness and overlap.
- Retain FID when it is part of the genotype key.
- Resolve repeated phenotype observations before regression.

Fictitious IDs and values. Reading connection: Chapter 8, section 8.3.2.

Today's small dataset

600 fictitious people

2,400 synthetic variants

A quantitative trait, age, a binary covariate, and five supplied PCs.

Deliberately introduced problems give QC something to find.

One planted association gives us a result to interpret.

All displayed records and results are synthetic.

QC asks questions at different levels

Level	Example question	Lab action
Site	Did this variant fail upstream filtering?	<code>-var-filter</code>
Sample	How many genotypes are missing for this person?	<code>-mind 0.05</code>
Variant	How many people lack a call here?	<code>-geno 0.02</code>
Allele	Do enough copies support this test?	<code>-maf 0.05</code> <code>-mac 20</code>

These are exercise thresholds. A study needs its own QC protocol.

Sources: supplied Chapter 8, sections 8.4–8.5; PLINK 2 input filtering.

The anatomy of a PLINK command

```
plink2 \  
  --pfile data/toy \  
  --freq --missing \  
  --threads 2 --memory 1024 \  
  --out results/raw
```

-pfile	input prefix, without the extension
-freq -missing	requested reports
-threads -memory	compute settings for this small exercise
-out	output prefix, shared by reports and log

Descriptive reports before exclusions

```
head -n 5 results/raw.afreq  
head -n 5 results/raw.smiss  
head -n 5 results/raw.vmiss  
less results/raw.log
```

.afreq	allele frequencies and observed allele counts
.smis	sample missingness
.vmiss	variant missingness
.log	inputs, options, counts, and warnings

Question: which report tells you about a poorly measured person?

Source: PLINK 2 file-format reference.

QC in explicit stages

```
plink2 --pfile data/toy --autosome --snps-only just-acgt \  
  --max-alleles 2 --var-filter --make-pgen --out results/sites  
plink2 --pfile results/sites --mind 0.05 \  
  --make-pgen --out results/samples  
plink2 --pfile results/samples --geno 0.02 --maf 0.05 \  
  --mac 20 --make-pgen --out results/qc
```

Stage	Samples	Variants
Input	600	2,400
Eligible sites	600	2,390
Sample filter	588	2,390
Variant filters	588	2,330

Synthetic results from the supplied package. Lab script adds explicit memory and thread settings.

QC also depends on the study population

- A pooled Hardy–Weinberg test can respond to population mixture.
- Heterozygosity and missingness need context from ancestry and technical batches.
- Related samples require an appropriate selection or modeling strategy.
- Rare variants need methods and allele-count rules suited to the question.

The lab omits a pooled HWE exclusion. We discuss what a real protocol must add.

Reading connection: Chapter 8, section 8.5. See also PLINK 2 filtering and association documentation.

Population structure and relatedness

Principal components

Summarize major patterns of genotype variation.

Use genome-wide, suitable markers or release-provided PCs for a real study.

LD pruning can reduce over-weighting of correlated markers.

Family relationships

Can leave residuals correlated even after PC adjustment.

Use a justified unrelated subset or a method that models relatedness.

A short genomic interval cannot support a full ancestry or kinship assessment.

Sources: PLINK 2 PCA, LD, and association documentation; AoU genomic data organization.

The association model

$$y_i = \alpha + \beta_j G_{ij} + \gamma \text{age}_i + \delta \text{sex}_i + \sum_{k=1}^5 \theta_k \text{PC}_{ik} + \varepsilon_i$$

y_i	quantitative phenotype in its original units
G_{ij}	ALT allele count at variant j in this exercise
β_j	conditional mean difference per additional ALT copy
Covariates	variables specified before looking at significant results

PLINK fits a separate regression at each variant.

A covariate-adjusted association scan

```
plink2 --pfile results/qc \  
  --pheno data/phenotypes.tsv --pheno-name trait \  
  --covar data/covariates.tsv \  
  --covar-name age,sex,PC1,PC2,PC3,PC4,PC5 \  
  --covar-variance-standardize age PC1 PC2 PC3 PC4 PC5 \  
  --glm hide-covar omit-ref \  
  --out results/assoc
```

hide-covar keeps covariate estimates out of the report.

omit-ref makes ALT the tested allele at these biallelic sites.

The report is results/assoc.trait.glm.linear.

Source: PLINK 2 association documentation.

Reading one association result

ID	A1	N	BETA	SE	P
SIMV001201	G	587	0.928	0.0626	2.19×10^{-42}

- One more G copy is associated with 0.928 more trait units, conditional on the included covariates.
- N is 587 here, although 588 samples passed QC.
- Check TEST=ADD and ERRCODE=. before interpreting P.
- For 2,330 tests, a simple Bonferroni threshold is $0.05/2330 \approx 2.15 \times 10^{-5}$.

Synthetic result. The deliberately planted effect was 0.9 per G copy.

Saving a reproducible run

```
# Fill config.sh with your own authorized workspace bucket.  
bash scripts/04_save.sh
```

The script packages the analysis code, configuration, results, logs, and input checksums, then copies the archive to the workspace bucket.

Verify the cloud copy before stopping the VM. Stop the app in the Workbench when finished.

With real AoU data: keep participant-level files and potentially sensitive outputs in the Workbench. Review dissemination rules before sharing results.

Sources: Verily Workbench storage guidance; AoU publication and dissemination policies.

Compliance can be a default, not a memory test

Dissemination policy: no participant count of 1–20 may be published, **including counts derivable from other reported numbers**. Zero is fine.

`scripts/check_disclosure.py` screens results/ inside every save. Today it catches two:

- `qc_counts.tsv`: 600 and 588 samples together reveal “12 removed”.
- `top_hits.tsv`: Ns of 587 and 588 reveal one person’s missing call.

In AoU mode a finding **blocks the save** until fixed or acknowledged after review.

Automate the checkable rules; keep human review for text, figures, percentages.

Source: AoU Data and Statistics Dissemination Policy. The screen is deliberately conservative; GWAS summary statistics have their own accepted practices.

Test the plumbing before you pay for a VM

A local sandbox mirrors the *shapes* of All of Us: layout, file names, headers, ID conventions, Workbench environment variables. Fake data, real bugs.

```
source ~/aou-sandbox/aou_sandbox/env.sh    # laptop: paths go local
bash scripts/00_preflight.sh    # ... 02, 03, and 04_save.sh
# on the Workbench: the same commands, unchanged
```

- Even the cloud save runs, into a fake local bucket, via a gcloud shim.
- Catches path, ID-join, FID-convention, and format bugs in seconds, free.
- Proves nothing about memory, runtime, statistics, or CDR queries.

Find your bugs on your laptop. Spend VM time on science, not typos.

Sandbox by Jon Jala. Twelve known AoU quirks are reproduced on purpose; see its traps.tsv.

Your 20-minute lab

-
- 40–43 Enter the workshop directory. Run preflight.
 - 43–48 Run QC. Explain the changes in sample and variant counts.
 - 48–54 Run association. Identify the counted allele and effect.
 - 54–58 Explain one missing observation and the multiple-testing threshold.
 - 58–60 Save the run. Complete the exit check.
-

```
bash scripts/00_preflight.sh
bash scripts/02_qc.sh
bash scripts/03_association.sh
bash scripts/04_save.sh
```

Questions to answer while your code runs

1. How many samples and variants survived QC? Which report records this?
2. At the leading variant, which allele does BETA refer to?
3. Why can the regression N differ from the number of samples after QC?
4. Does a p-value of 0.001 pass our correction for 2,330 tests?
5. Where is your saved run, and could you reconstruct its inputs?

Optional extension: change one QC threshold and explain the consequences before rerunning.

The exit check

“I can locate my inputs, explain my filters,
interpret one effect, and recover my run.”

For your next project

Choose a phenotype and a small, approved genomic subset.

Write down the release, sample selection, QC, and model before scaling.

Ask whether the method handles the relatedness and diversity in your actual sample.

“It’s astonishing that this frenzied molecular party maintains enough order to keep life viable at all.”

— Philip Ball, *How Life Works* (2023)

Appendix: preparing a real AoU teaching subset

1. In a Controlled Tier workspace, select one CDR release and read its genomic directory and QC report.
2. Define one phenotype per person, units, observation window, and missingness rules.
3. Prepare a small autosomal genotype slice with a justified sample selection. Apply the release's sample and genotype filters.
4. Join matching genome-wide PCs and phenotype/covariate records by IDs. Resolve relatedness before using ordinary regression.
5. Stage the six-file bundle specified in docs/AOU_ADAPTATION.md. Rehearse on the participants' actual VM image.

Preparation happens before class. Participants analyze the prepared slice.

Sources: AoU genomic analysis guidance; release-specific data organization and QC documents.

Jeff Swigert USC CESR All of Us CLI workshop

Appendix: a small BigQuery connection check

After selecting the dataset and billing project in your workspace:

```
# CDR_DATASET is the verified project.dataset reference.
bq --project_id="$BILLING_PROJECT" query \
  --use_legacy_sql=false --maximum_bytes_billed=10000000 \
  "SELECT column_name, data_type
  FROM \`${CDR_DATASET}.INFORMATION_SCHEMA.COLUMNS`
  WHERE table_name = 'measurement'"
```

This inspects the schema. A phenotype query still needs clinical definitions, concept selection, units, dates, and a one-row-per-person rule.

Sources: AoU Data Dictionaries; Google BigQuery INFORMATION_SCHEMA.COLUMNS and bq CLI documentation.

Appendix: a real phenotype, live

From browser to script in three steps.

1. **Data Browser** (databrowser.researchallofus.org, public): search the trait; read off the **concept ID**, units, and counts.
2. Confirm domain and source-vs-standard concept in the **Data Dictionaries**. Cohort/Dataset Builders can write the SQL for you.
3. Paste the ID into a parameterized script; query aggregates only:

```
bash scripts/06_pheno_summary.sh          # the menu
bash scripts/06_pheno_summary.sh height  # one real summary
```

The menu spans program measurements, EHR labs, surveys, and conditions, each with its source type's trap. The script prints **resolved concept names first** — an ID is a claim until the name confirms it — byte-caps queries, and suppresses counts of 1–20.

Requires Controlled Tier and CDR_DATASET + BILLING_PROJECT in config.sh. IDs distilled from a validated pipeline; re-verify per CDR release.

Appendix: LD pruning and PCA on the toy data

```
plink2 --pfile results/qc \  
  --indep-pairwise 200kb 0.2 --out results/prune  
plink2 --pfile results/qc \  
  --extract results/prune.prune.in \  
  --pca 5 --out results/pca
```

.prune.in lists the retained markers.

.eigenvec gives each sample's PC coordinates.

For real AoU analyses, use a suitable genome-wide marker panel or the release's supplied PCs. Recomputing PCs on a narrow interval changes what they measure.

Sources: PLINK 2 LD and population stratification documentation.

Appendix: common problems and their first checks

Problem	First check
command not found	Confirm PLINK 2 path and the instructor's installed build.
No such file	Use <code>pwd</code> , then <code>ls</code> ; verify all three genotype files.
0 samples or phenotypes	Check ID headers, FID/IID encoding, duplicates, and overlap.
All associations are NA	Inspect <code>ERRCODE</code> , covariate variance, collinearity, and MAC.
403 / access denied	Confirm selected workspace, data access, and billing context.
Killed / disk full	Inspect memory and disk use; shrink the subset before scaling.

Appendix: how the reading maps to the workshop

Reading	Workshop connection
Chapter 7, section 7.5	Recognizing VCF and PLINK file families
Chapter 7, section 7.7	Storage, transfer, and compute planning
Chapter 8, section 8.2	Shell commands, PLINK calls, and logs
Chapter 8, sections 8.3–8.4	Subsetting, attaching phenotypes, and descriptive reports
Chapter 8, section 8.5	Sample QC, variant QC, and analysis-specific decisions

Mills, M. C., Barban, N., and Tropf, F. C. (2020). An Introduction to Statistical Genetic Data Analysis. MIT Press. Supplied Chapters 7 and 8.

Appendix: platform references

- AoU: CDR version 9 announcement
- AoU: Data Dictionaries and current resource references
- AoU: Genomic and multi-omic data organization
- AoU: Getting started with genomic analyses
- Verily: JupyterLab apps and file persistence
- AoU: Publication, presentation, and dissemination checklist

Checked 18 September 2026. Use the documentation for the release and environment in your workspace.

Appendix: software and methods references

- PLINK 2: downloads and version history
- File formats Input filtering Association analysis
- LD pruning Principal components
- BigQuery column metadata bq CLI reference
- Mills, Barban, and Tropf (2020), Chapters 7–8, supplied readings.

Design: USC Cardinal #990000 and USC Gold #FFCC00, following USC color guidance.